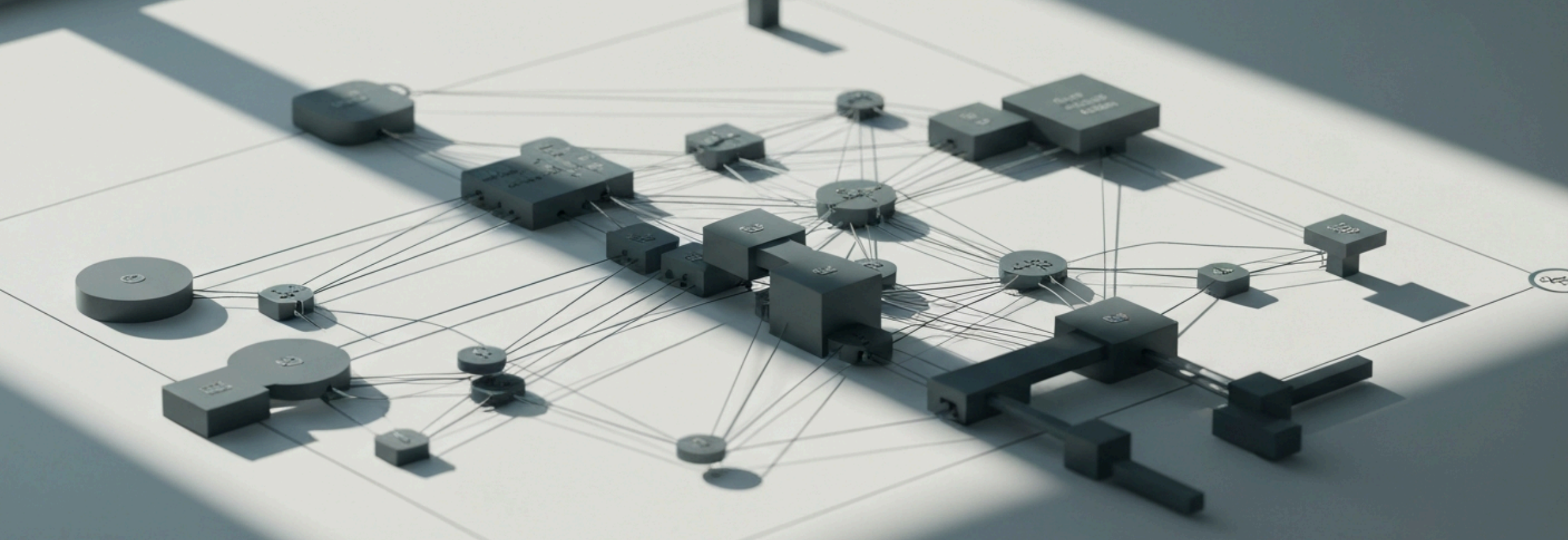
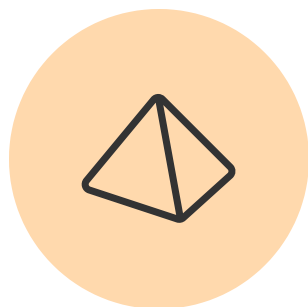




Implementing RAG Graphs to Enhance LLM Capability

Matthew, Mridul, Arsam

Introduction



Proliferation of misinformation and fake news

Difficulty for users to determine trustworthiness of news articles and claims



Limitations of current LLMs

Unreliable responses, hallucinated information, lack of grounding in evidence



Critical need for verified information

Requirement for systems that can provide transparent justifications backed by credible sources

Developing a system that can verify information and provide clear, evidence-backed explanations is crucial to address the growing challenge of misinformation and lack of trust in current LLM outputs.

Dataset



100 articles from credible news sources

The dataset consists of 100 articles collected from well-established and reliable news organizations.



Articles published in 2025

The articles were published between January 1, 2025, and December 1, 2025.



Verified factual information

The articles contained verified factual information, ensuring the dataset was of high quality with minimal noise.



Structured list of short claims

Each article was transformed into a structured list of short, verifiable claims to enable efficient embedding, retrieval, and reasoning.

The dataset of 100 well-curated news articles, containing verified factual information, was a crucial component in the development of the proposed RAG-based system.

Data Preprocessing



FLAN-T5-base model used for claim extraction

This lightweight yet capable NLP model was well-suited for the text-to-text tasks required in this project due to its instruction-following capabilities and deterministic outputs.



Claim and Evidence Embeddings

Sentence Transformer model used to convert text into fixed-length numerical embeddings. Dimensionality reduced via PCA, then L2 normalized for efficient cosine similarity computations.



Preprocessed evidence stored with metadata

Unique identifiers, similarity and relevance scores computed at query time.

The data preprocessing steps, including the use of the FLAN-T5-base model for claim extraction and the storage of text as normalized embeddings with associated metadata, laid the groundwork for the effective implementation of the graph-based RAG system.

RAG Overview



RAG Framework Overview

Retrieval-Augmented Generation (RAG) framework improves reliability and transparency of LLMs by grounding outputs in verified evidence



Motivation

Address issues of misinformation and fake news by verifying news articles and claims using a knowledge base



Key Idea

Cluster verified information into semantically meaningful groups and retrieve clusters most relevant to user's query



Workflow

Extract and preprocess data, generate word representations, create and search graph, combine claim and evidence for LLM classification

The proposed RAG framework leverages a graph-based retrieval and multi-hop search approach to enhance the trustworthiness and transparency of LLM outputs by grounding them in credible evidence.

Graph-Based Clustering (k-Means++)

Evidence Embedding Creation

Use a pre-trained Sentence Transformer model to convert the evidence sentences into 768-dimensional numerical embeddings.

Dimensionality Reduction with PCA

Apply Principal Component Analysis (PCA) to compress the 768-dimensional embeddings to around 300 dimensions while retaining 90% of the variance. Then, apply L2 normalization to ensure efficient cosine similarity computations.

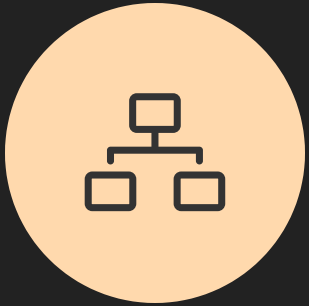
k-Means++ Clustering

Use the k-Means++ algorithm to partition the embedding space into k clusters, where each cluster represents a semantically related group of evidence sentences. k-Means++ improves upon standard k-Means by using a smarter initialization strategy for the cluster centroids, reducing the likelihood of poor local optima.

Evidence Graph Construction

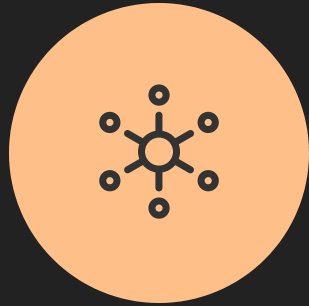
View the clustered structure of the evidence embeddings as a semantic graph, where each cluster corresponds to a node containing a group of related evidence sentences. The distances between the centroid vectors implicitly define relationships in the embedding space, which can be used for search and retrieval.

Word Embeddings & PCA



Sentence Transformer Model

Used sentence-transformers/all-MiniLM-L6-v2 encoder to generate 768-dimensional sentence embeddings



Semantic Similarity Capture

Transformer's attention mechanism assigns higher importance to key words, entities, and negations to preserve contextual meaning



Principal Component Analysis (PCA)

Applied PCA to compress 768-dimensional embeddings to 300 dimensions while retaining most of the variance and semantic information

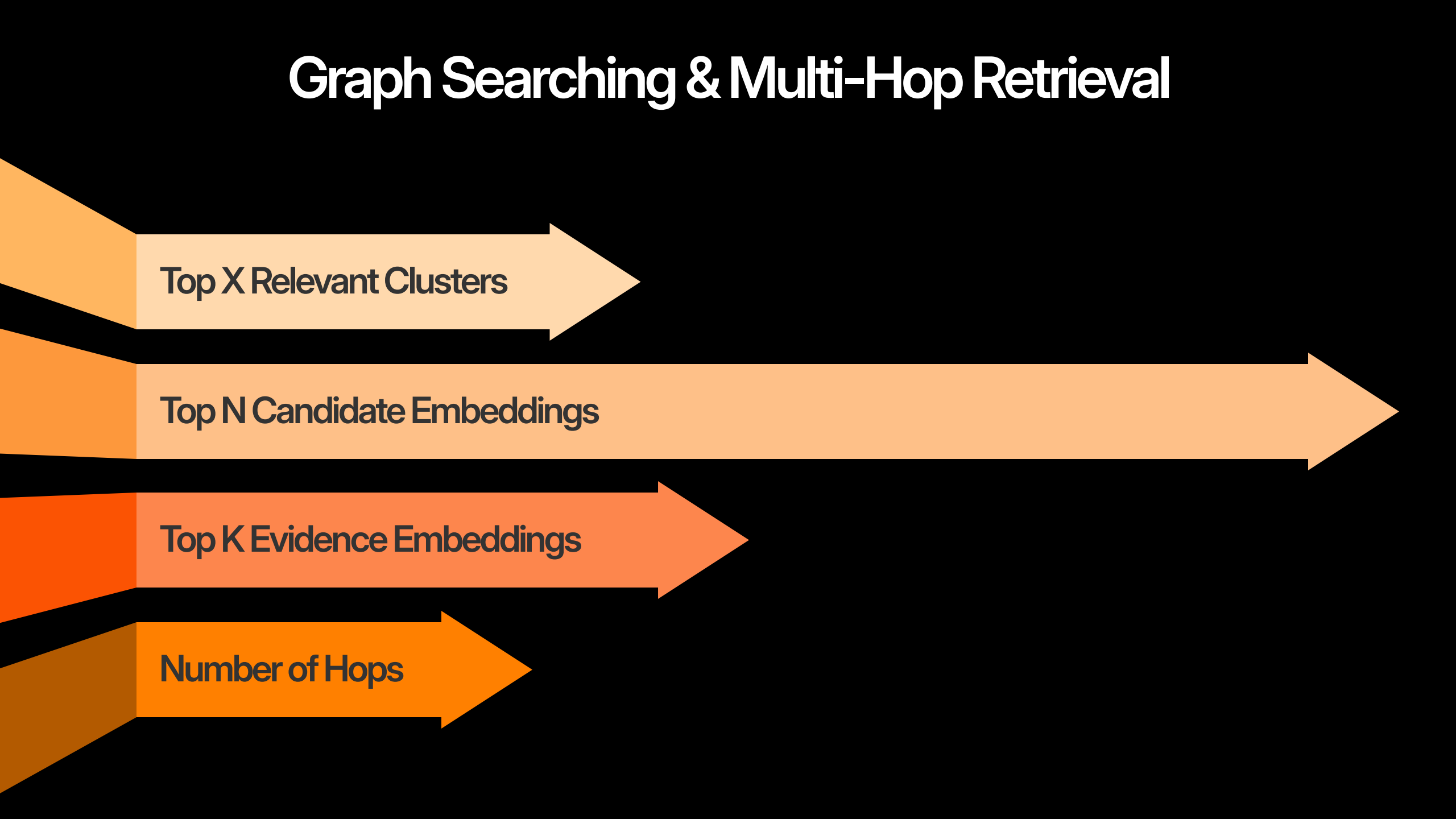


L2 Normalization

Normalized the embeddings to unit length to enable efficient cosine similarity computations during retrieval

The compressed, semantically-meaningful 300-dimensional embeddings allow for efficient similarity-based retrieval and reasoning to identify the most relevant facts for verifying the user's claim.

Graph Searching & Multi-Hop Retrieval



Top X Relevant Clusters

Top N Candidate Embeddings

Top K Evidence Embeddings

Number of Hops

RAG + LLM Pipeline

FLAN-T5-Base Model for claims

The system leverages the pre-trained FLAN-T5-base (250M parameters) model to extract claims from articles and adds them to a list.

Claim + Top K Evidence

The input claim is combined with the top K reranked evidence retrieved from the graph-based search, providing the necessary context for the language model to generate the final classification.

Generating Verification Output

The fine-tuned FLAN-T5-base model generates the fact-checked decision, along with a supporting justification based on the provided evidence, enhancing the transparency and trustworthiness of the system's outputs.

Hyper Parameter Configuration

TOP_X

3 clusters searched per query

TOP_N

10 candidates retrieved per hop

TOP_K

4 final reranked results

NUM_HOPS

2 for multi-hop retrieval

Number of Clusters

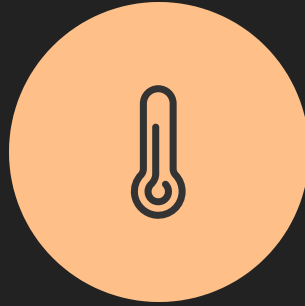
17 groups

Evaluation & Metrics



Tested on balanced FEVER dataset

300 claims supported by graph, 300
claims refuted by graph



Evaluated using standard classification metrics

Precision, recall, F1 score



Consistently high recall across all test cases

Likely due to model's conservative
approach of assuming claims are false
unless evidence proves otherwise

The team's model demonstrated strong performance on the balanced FEVER dataset, with consistently high recall across all test cases. This suggests the RAG-based approach effectively leverages the graph structure to provide reliable evidence for claim verification.

23% better with minimal training data

	precision	recall	f1-score	support
NOT ENOUGH INFO	0.00	0.00	0.00	0
REFUTES	0.77	0.72	0.75	162
SUPPORTS	0.88	0.55	0.68	138
accuracy			0.64	300
macro avg	0.55	0.42	0.48	300
weighted avg	0.82	0.64	0.72	300

RAG LLM

	precision	recall	f1-score	support
REFUTES	0.54	0.86	0.66	162
SUPPORTS	0.44	0.12	0.19	138
accuracy			0.52	300
macro avg	0.49	0.49	0.43	300
weighted avg	0.49	0.52	0.45	300

LLM Without RAG

Challenges



Insufficient Context in Knowledge Base

Input queries sometimes touched on topics that the model had no prior information on, leading to a lack of relevant evidence in the knowledge base.



Suboptimal Clustering of Evidence

The evidence embedding chunks were sometimes grouped in a way where vital information was spread across too many clusters or useless information was included, making it harder for the LLM to extract the correct evidence.



Challenges in Multi-Hop Retrieval

The multi-hop search process to find the best evidence was sometimes hindered by the lack of context and suboptimal clustering, making it difficult to identify the most relevant information.

The proposed RAG-based system faced challenges in maintaining a comprehensive knowledge base and efficiently organizing the evidence, which hindered the LLM's ability to reliably ground its outputs in credible information. Addressing these issues will be crucial for improving the system's overall performance.

Future Work

- **Expand dataset coverage**

Incorporate a larger and more diverse dataset, including verified news articles, factual statements, and information from multiple online sources

- **Explore alternative graph representations**

Investigate different ways to represent the semantic relationships between evidence sentences, such as using alternative graph structures or clustering algorithms

- **Experiment with different clustering methods**

Explore alternative clustering techniques beyond k-means++ to better capture the nuanced semantic structure of the evidence corpus

- **Improve evidence retrieval**

Enhance the graph searching and multi-hop retrieval process to ensure the most relevant and coherent evidence is retrieved

- **Enhance LLM grounding**

Further refine the integration of the RAG framework with the language model to improve the trustworthiness and transparency of the final outputs

References

- **2018 – Yang et al., "TI-CNN: Convolutional Neural Networks for Fake News Detection."**
- **2021 – Nasir et al., "Fake News Detection: A Hybrid CNN–RNN Deep Model"**
- **2024 – Chang et al., "Graph Global Attention Network with Memory (GANM)"**
- Lewis, P., Oguz, B., Rinott, R., Riedel, S., & Schwenk, H. (2020).
- Raffel, C. et al. (2020).